TITLE:

Why Oil & Gas EPC Projects Get Delayed10 Critical Schedule Risks and Proven Mitigation Strategies

Athor:

Behzad Mousavi, MSc in Industrial Engineering
Planning & Project Control Specialist

Executive Summary

Engineering, Procurement, and Construction (EPC) projects in the oil and gas industry are among the most complex and capital-intensive projects in the world. They involve thousands of engineering deliverables, extensive procurement activities, multidisciplinary construction works, strict quality standards, and demanding commissioning requirements.

Despite the widespread adoption of advanced scheduling software such as Primavera P6 and Microsoft Project, schedule overruns remain one of the most significant challenges facing EPC contractors and project owners. Delays not only increase direct construction costs but also postpone production start-up, reduce return on investment, create contractual disputes, and negatively affect stakeholder confidence.

This white paper identifies the ten most common causes of schedule delays in oil and gas EPC projects and presents practical mitigation strategies based on internationally recognized project control practices. The objective is to help project managers, planners, schedulers, construction managers, and project control engineers improve schedule reliability and achieve successful project delivery.

About the Author

Behzad Mousavi is a Project Planning & Control Specialist with more than twenty-five years of experience in industrial EPC projects, particularly within the oil and gas sector. Throughout his career, he has been actively involved in engineering coordination, procurement planning, construction scheduling, delay analysis, recovery planning, project controls, and Primavera P6 implementation.

His professional experience spans all major project phases—from Front-End Engineering Design (FEED) to commissioning and start-up—providing him with a comprehensive understanding of the challenges associated with delivering complex industrial projects.

Introduction

Oil and gas EPC projects are inherently challenging due to their multidisciplinary nature, large-scale investments, international supply chains, and stringent contractual requirements. Every project phase—from engineering and procurement to construction and commissioning—is interconnected, meaning that delays in one area often propagate throughout the entire project lifecycle.

Although sophisticated planning tools have significantly improved schedule development and monitoring capabilities, many projects continue to experience substantial delays. In most cases, these delays are not caused by a single issue but rather by a combination of engineering deficiencies, procurement bottlenecks, resource limitations, ineffective communication, and inadequate project controls.

Successful organizations recognize that schedule management extends far beyond creating a baseline schedule. It requires continuous monitoring, proactive risk management, accurate forecasting, and timely decision-making throughout the project lifecycle.

This paper explores the ten most critical causes of schedule delays and provides practical recommendations that project teams can implement to improve schedule performance and reduce project risks.

---

1. Incomplete Engineering Deliverables

Engineering forms the backbone of every EPC project. High-quality engineering deliverables provide the technical foundation upon which procurement, fabrication, construction, and commissioning activities depend. When engineering outputs are delayed, incomplete, or frequently revised, downstream project activities inevitably suffer.

Common engineering-related issues include:

- **Late issuance of IFC (Issued for Construction) drawings**
- **Frequent design revisions**
- **Delayed vendor document approvals**
- **Incomplete material specifications**
- **Missing interdisciplinary interfaces**
- **Poor document control**
- **Inconsistent engineering priorities**

Even seemingly minor engineering delays can trigger a cascading effect across procurement, fabrication, logistics, and construction activities. Contractors often mobilize resources based on engineering milestones; therefore, delayed deliverables frequently result in idle manpower, equipment downtime, and increased project costs.

Best Practice

Establish Engineering Progress Curves based on weighted deliverables rather than document counts. Conduct weekly engineering performance reviews and closely monitor engineering milestones against the approved baseline schedule.

Expert Tip

Engineering progress should always be measured based on deliverable weight and maturity rather than the total number of issued documents.

---

2. Delays in Procuring Long-Lead Equipment

Procurement is one of the most schedule-sensitive phases of an EPC project. Many critical equipment packages require extended manufacturing periods, factory acceptance testing (FAT), international transportation, customs clearance, and site inspection before installation can begin.

Typical long-lead items include:

- Gas Turbines
- Compressors
- Pressure Vessels
- Heat Exchangers
- Large Power Transformers
- Reactors
- Major Pumps

A delay in manufacturing or shipment of any critical equipment package can postpone construction completion, mechanical completion, and ultimately project commissioning.

The procurement team must therefore maintain close coordination with engineering, vendors, logistics providers, and construction management to ensure timely delivery.

Best Practice

Identify all Long-Lead Items during the FEED stage and establish dedicated procurement tracking reports. Monitor manufacturing progress through regular vendor meetings, expediting activities, and logistics reviews.

Common Mistake

Many project schedules focus on purchase order dates while neglecting manufacturing milestones, inspection activities, and international shipping durations.

---

3. Ineffective Interface Management

Large EPC projects involve multiple engineering disciplines, contractors, subcontractors, vendors, consultants, and client representatives working simultaneously. Without effective interface management, coordination problems quickly emerge and disrupt project execution.

Typical interface conflicts include:

- Civil versus Mechanical installation
- Mechanical versus Electrical works
- Electrical versus Instrumentation systems
- Construction versus Commissioning activities
- Vendor responsibilities versus Contractor scope
- Temporary facilities versus permanent installations

Poor interface coordination often results in work stoppages, redesign, rework, resource conflicts, and contractual disputes.

An effective Interface Management Plan clearly defines responsibilities, communication procedures, approval workflows, and issue resolution mechanisms between all project participants.

Best Practice

Conduct multidisciplinary interface coordination meetings every week. Maintain an Interface Register to monitor unresolved issues and assign responsibility for timely resolution.

Project Control Insight

Interface management should be treated as a continuous project control activity—not merely a coordination meeting. Unresolved interfaces frequently become hidden drivers of critical path delays.

4. Scope Changes and Their Impact on Project Schedules

Scope changes are inevitable in most EPC projects. Even with comprehensive Front-End Engineering Design (FEED) studies, unforeseen technical requirements, client requests, regulatory updates, or operational improvements may require modifications during project execution.

While some changes are essential to improve safety, reliability, or operational efficiency, uncontrolled scope growth—commonly known as *scope creep*—is one of the leading causes of schedule overruns and cost escalation.

Typical sources of scope changes include:

- Additional client requirements
- Process design modifications
- Equipment substitution or upgrades
- Regulatory and environmental compliance updates
- Changes resulting from HAZOP or constructability reviews
- Site condition discoveries
- Vendor design revisions

Every approved change has consequences that extend beyond the engineering department. A seemingly minor modification can require redesign, new procurement activities, fabrication changes, construction rework, revised testing procedures, and updated commissioning plans.

Without systematic evaluation, the cumulative impact of numerous small changes can significantly delay project completion.

Best Practice

Implement a formal **Change Management Process** integrated with schedule control, cost control, document management, and risk management. Every approved change should be accompanied by a documented Schedule Impact Analysis (SIA) before implementation.

Expert Tip

The greatest risk is not a single large change—it is the accumulation of many small changes whose combined impact remains invisible until the project is already behind schedule.

---

5. Resource Constraints

Construction productivity is directly influenced by the availability of qualified resources. Even the most accurate project schedule cannot be achieved if the required workforce, equipment, materials, or financial resources are unavailable when needed.

Resource shortages frequently occur because of poor planning, inaccurate forecasting, labor market limitations, or competing projects.

Typical resource constraints include:

- Skilled labor shortages
- Insufficient construction equipment
- Limited site supervision
- Material delivery delays
- Fabrication bottlenecks
- Financial constraints affecting procurement
- Limited access to specialized subcontractors

Resource-related delays rarely remain isolated. A shortage in one discipline often creates a domino effect that disrupts multiple downstream activities and reduces overall site productivity.

Advanced planning tools such as Primavera P6 allow project teams to perform resource loading, leveling, and forecasting, helping identify shortages before they become critical.

Best Practice

Perform monthly resource-loading reviews comparing planned versus actual manpower, equipment utilization, and material availability. Develop early mitigation plans whenever significant deviations are identified.

Project Control Insight

Monitoring manpower numbers alone is insufficient. Productivity trends, workforce efficiency, and discipline-specific performance indicators provide a far more reliable measure of project progress.

---

6. Weak Schedule Updating Practices

Many organizations believe that updating activity progress percentages constitutes effective schedule control. In reality, schedule updating is only the starting point of project performance analysis.

A schedule update should provide management with meaningful insights into current performance, future risks, and potential recovery opportunities—not simply report completed activities.

A comprehensive schedule review should evaluate:

- Critical Path evolution
- Near-Critical activities
- Float consumption trends
- Schedule Variance (SV)
- Forecast Completion Date
- Milestone performance
- Progress against baseline
- Driving relationships
- Resource utilization
- Potential schedule risks

Without proper analysis, schedule updates become administrative reporting exercises rather than management tools.

Project managers require actionable information—not merely updated Gantt charts.

Best Practice

After every schedule update, perform a complete Critical Path Analysis and compare current project logic with the approved baseline schedule. Investigate all changes affecting project completion dates or critical milestones.

Common Mistake

Updating progress percentages without analyzing schedule logic often creates a false impression that the project is under control.

---

7. Inadequate Risk Management

Every EPC project operates within an environment of uncertainty. Political events, economic fluctuations, technical challenges, environmental conditions, and supply chain disruptions all introduce risks capable of affecting project schedules.

Organizations that treat risk management as a one-time planning exercise often struggle to respond effectively when unexpected events occur.

Major schedule risks may include:

- Political sanctions

- Import restrictions
- Extreme weather conditions
- Currency fluctuations
- Global supply chain disruptions
- Labor strikes
- Safety incidents
- Regulatory approvals
- Vendor insolvency
- Transportation delays

Risk management should not focus solely on identifying threats. It must also define practical mitigation measures, contingency plans, and clear ownership for each identified risk.

Risk registers should remain active throughout the project lifecycle and be reviewed regularly as project conditions evolve.

Best Practice

Maintain a dynamic Risk Register integrated with project scheduling. Regularly assess probability, impact, mitigation effectiveness, and residual risk, and ensure high-priority risks are reflected in schedule reviews and management reporting.

Expert Tip

Effective risk management is proactive rather than reactive. The objective is not merely to document risks—it is to reduce their probability and minimize their impact before they affect project execution.

8. Poor Communication and Delayed Decision-Making

Technical expertise alone cannot guarantee project success. In many EPC projects, schedule delays are not caused by engineering or construction challenges but by slow decision-making and ineffective communication between stakeholders.

Large projects involve owners, EPC contractors, subcontractors, engineering consultants, equipment manufacturers, inspection agencies, and regulatory authorities. Without clear communication channels, approvals become delayed, technical questions remain unresolved, and construction teams are often forced to suspend activities while awaiting decisions.

Common communication issues include:

- Slow approval of engineering documents
- Delayed responses to Requests for Information (RFIs)
- Inefficient document distribution
- Poor coordination between contractors and vendors

- Unclear responsibilities among project stakeholders
- Late management decisions affecting procurement or construction

These issues may appear minor individually, but together they can significantly affect the project's critical path and overall schedule performance.

Best Practice

Establish a structured communication framework that defines reporting lines, approval authorities, document turnaround times, and escalation procedures. Weekly coordination meetings should focus on unresolved issues requiring immediate management decisions.

Project Control Insight

Projects rarely fail because information is unavailable. They fail because critical information does not reach the right decision-makers at the right time.

---

9. Unrealistic Baseline Schedules

A project schedule is only as reliable as the assumptions on which it is built.

Unfortunately, many EPC projects begin with overly optimistic baseline schedules developed primarily to satisfy commercial negotiations or contractual commitments rather than reflecting realistic project conditions.

Typical weaknesses include:

- Underestimated engineering durations
- Unrealistic procurement lead times
- Overestimated construction productivity
- Ignoring seasonal weather impacts
- Inadequate resource planning
- Insufficient schedule contingency

When the original baseline is unrealistic, project teams spend much of the execution phase explaining delays rather than preventing them.

A realistic baseline schedule should reflect achievable productivity rates, validated procurement durations, engineering maturity, and available resources.

Best Practice

Validate the baseline schedule through multidisciplinary reviews before project approval. Experienced planners, engineering managers, procurement specialists, construction managers, and project controls personnel should jointly verify schedule logic, durations, and resource assumptions.

Common Mistake

An unrealistic schedule may look attractive during contract negotiations but often becomes the primary source of project performance problems during execution.

---

10. Lack of Recovery Planning

No major EPC project progresses exactly as planned. Delays are inevitable; however, successful organizations distinguish themselves by how rapidly and effectively they respond to emerging schedule slippages.

Recovery planning should not begin when a project is already significantly behind schedule. Instead, it should become an integral part of ongoing schedule management.

Typical recovery techniques include:

- Fast Tracking
- Schedule Crashing
- Additional Work Shifts
- Resource Reallocation
- Construction Resequencing
- Parallel Engineering Activities
- Prefabrication and Modular Construction
- Prioritizing Critical Path Activities

Each recovery option should be evaluated for its impact on project cost, quality, safety, and contractual obligations.

Best Practice

Develop multiple recovery scenarios before schedule slippage reaches critical levels. Regular schedule risk reviews enable project teams to activate recovery plans early, minimizing disruption and preserving project objectives.

Expert Tip

Recovery planning is most effective when initiated proactively. Waiting until major delays occur usually limits available recovery options and increases overall project costs.

The Strategic Role of Primavera P6

Primavera P6 is far more than a scheduling application—it is a comprehensive project management and decision-support platform.

When implemented effectively, Primavera P6 enables project teams to monitor progress, analyze schedule performance, identify emerging risks, evaluate recovery scenarios, and forecast project completion with greater confidence.

Project planners should routinely analyze:

- Critical Path
- Near-Critical Activities
- Float Consumption
- Schedule Performance Index (SPI)
- Earned Value Metrics
- Forecast Completion Dates
- Resource Utilization
- Milestone Trends
- Schedule Variance
- Recovery Opportunities

Rather than simply updating activity percentages, experienced planners use Primavera P6 to generate meaningful insights that support proactive management decisions.

The true value of Primavera P6 lies not in producing Gantt charts, but in providing timely information that enables project teams to anticipate problems before they become critical.

Conclusion

Schedule delays in oil and gas EPC projects rarely result from a single isolated issue. Instead, they are typically the consequence of multiple interconnected factors involving engineering, procurement, construction, communication, resource management, and project controls.

Organizations that consistently deliver successful projects recognize that schedule management extends far beyond developing a baseline schedule. It requires continuous monitoring, disciplined performance measurement, proactive risk management, structured communication, and timely recovery planning.

Modern project controls should focus not merely on reporting historical performance but on forecasting future outcomes and enabling informed decision-making. By integrating robust planning practices with advanced tools such as Primavera P6, organizations can significantly improve schedule reliability, reduce project risks, and enhance overall project performance.

Ultimately, successful project control is not measured by the number of reports produced—it is measured by the quality of decisions those reports enable.

Key Takeaways

- High-quality engineering deliverables are essential for successful project execution.
- Long-lead procurement activities should be identified and monitored from the earliest project phases.
- Effective interface management minimizes conflicts between disciplines and contractors.
- Schedule updates must include comprehensive Critical Path and float analysis.
- Active risk management should remain integrated throughout the entire project lifecycle.
- Resource planning must focus on both availability and productivity.
- Recovery planning should begin before schedule slippage becomes critical.
- Primavera P6 delivers the greatest value when used as a strategic project control and decision-support system rather than merely as scheduling software.

About PMCenter

PMCenter is dedicated to advancing excellence in Project Planning and Control through high-quality technical publications, professional training, and practical industry guidance.

Our mission is to help project professionals improve planning accuracy, strengthen schedule management, and successfully deliver complex industrial projects.

Areas of Expertise

- Primavera P6 Professional
- Project Planning & Scheduling
- Project Controls
- Delay Analysis
- Recovery Planning
- Critical Path Method (CPM)
- Earned Value Management (EVM)
- Risk Management
- Engineering, Procurement & Construction (EPC)
- Oil & Gas Project Management

For more technical articles, training materials, and professional resources, visit our page in GitHub as a link below:

https://sbmousavices.github.io/industrial-Tools